

$$\sum_{n=0}^{\infty} n u(n) \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} n z^{-n}$$

$$= \frac{1}{z} + \frac{2}{z^2} + \dots + \frac{n}{z^n} = S(n)$$

$$\text{then } zS(n) = 1 + \frac{2}{z} + \frac{3}{z^2} + \dots + \frac{n}{z^{n-1}}$$

$$zS(n) - S(n) = 1 + \frac{1}{z} + \frac{1}{z^2} + \dots + \frac{1}{z^{n-1}} - \frac{n}{z^n}$$

$$= \frac{1-z^{-n}}{1-z^{-1}} - \frac{n}{z^n}$$

$$= \frac{1}{1-z^{-1}} \quad (n \rightarrow \infty)$$

$$\Rightarrow S(n) = \frac{z}{(z-1)^2}$$

$$\sum_{n=0}^{\infty} n^2 u(n) \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} n^2 \cdot z^{-n}$$

$$= \frac{1}{z} + \frac{2^2}{z^2} + \frac{3^2}{z^3} + \dots + \frac{n^2}{z^n} = S(n)$$

$$\text{then } zS(n) = 1 + \frac{2^2}{z} + \frac{3^2}{z^2} + \dots + \frac{n^2}{z^{n-1}}$$

$$zS(n) - S(n) = 1 + \frac{2^2-1^2}{z} + \frac{3^2-2^2}{z^2} + \dots + \frac{n^2-(n-1)^2}{z^{n-1}} - \frac{n^2}{z^n}$$

$$= 1 + \frac{3}{z} + \frac{5}{z^2} + \dots + \frac{2n-1}{z^{n-1}} - \frac{n^2}{z^n} = E(n)$$

$$\text{then } zE(n) = z + 3 + \frac{5}{z} + \frac{7}{z^2} + \dots + \frac{2n-1}{z^{n-2}} - \frac{n^2}{z^{n-1}}$$

$$zE(n) - E(n) = z + 2\left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots + \frac{1}{z^{n-2}}\right) - \frac{zn^2}{z^n} + \frac{n^2}{z^n}$$

$$= z + 2 \frac{1}{1-z^{-1}} \quad (n \rightarrow \infty)$$

$$= \frac{2z + z^2 - z}{z-1}$$

$$= \frac{z(z+1)}{z-1}$$

$$\Rightarrow E(n) = \frac{z(z+1)}{(z-1)^2}$$

$$\Rightarrow S(n) = \frac{z(z+1)}{(z-1)^3}$$

$$\Rightarrow \sum_{n=-\infty}^{\infty} n^2 u(n) \cdot z^{-n} = \frac{z(z+1)}{(z-1)^3} \quad (z > 1)$$